In silico construct design (His-tag placement and disulphide-bridge mapping)

What you're going to get / What you expect

For each candidate enzyme, a decision on the position of the polyhistidine tag (and therefore the expression vector) and a map of its structural disulphide bridges, all before any wet-lab work.

The 6xHis tag must sit on a solvent-exposed terminus so it can reach the Ni-NTA resin during purification without occluding the active site during catalysis. The more exposed terminus therefore decides the vector.

A structural disulphide bond will not form in the reducing cytoplasm of a standard *E. coli* strain, so a disulphide-dependent enzyme expressed there tends to misfold and accumulate as inclusion bodies. Detecting these bonds in advance tells you to express that enzyme in an oxidising-cytoplasm strain (for example Rosetta-gami 2, SHuffle T7 or Origami 2), planning the folding strategy before the first expression round instead of discovering it from a failure.

Materials list (per enzyme)

- Amino acid sequence of each enzyme (FASTA).
- SWISS-MODEL server (homology modelling for enzymes without a resolved structure).
- PyMOL (visualisation, labelling and distance measurement).
- pET-28a(+) and pET-22b(+) vector maps.

Steps

1. Compile each sequence in FASTA and, for enzymes without a resolved structure, build a homology model on SWISS-MODEL; open the model in PyMOL.
2. Run the protein-ends script (below) to colour the chain from N- to C-terminus and mark both ends; read off which terminus is more solvent-exposed.
3. Assign the vector from the exposed terminus: pET-28a(+) for an N-terminal tag (NdeI site in 5' plus STOP codon) or pET-22b(+) for a C-terminal tag.
4. Run the disulphide-bridge script (below) to detect pairs of cysteine sulphur (SG) atoms within a 2.5 A cutoff.
5. Route each disulphide-dependent enzyme to an oxidising-cytoplasm strain for later expression; record, per enzyme, the predicted mass, oligomeric state, tag position, vector and number of disulphide bridges.

PyMOL script - location of the protein ends

```
# 1. Clean up and set base view
Hide all
show cartoon, all
Gray80 color, all
# 2. Isolate one of the monomers if it is multihomomeric
# Select atom 1 of the file and its entire chain.
select monomero1, bychain (first all)
# 3. Apply color gradient to that chain
Spectrum Count, Rainbow, Monomer1
# 4. Locate and mark the N-terminal
select n_term, first (name CA and monomero1)
show spheres, n_term
Color Blue, n_term
label n_term, "N-Term"
# 5. Locate and mark the C-terminal of monomer 1
select c_term, last (name CA and monomero1)
show spheres, c_term
Color Red, c_term
label c_term, "C-Term"
# 6. Configure the typography of the labels (black, large and separate)
Set label_size, 30
Set label_color, Black
Set label_position, [5, 5, 5]
# 7. Set the transparent background for export and center the view
bg_color white
Set ray_opaque_background, 0
orient
```

PyMOL script - determination of disulphide bridges (2.5 A cutoff)

```
# 1. Clean up and set base view
Hide all
show cartoon, all
Gray80 color, all
# 2. Selection of all cysteines
select cys_all, resn CYS
show sticks, cys_all
Color Purple, cys_all
# 3. Highlight sulphur (SG) atoms
select sulphurs, resn CYS and name SG
show spheres, sulphurs
Set sphere_scale, 0.3, sulphurs
tv_purple color, sulphurs
# 4. Look for distances between atoms of the same type using a 2.5 A cutoff
dist puentes_ss, sulphurs, sulphurs, mode=2, cutoff=2.5
# 5. Visual configuration of the bridges as dashed lines
Set dash_color, Purple
Set dash_radius, 0.1
Set dash_length, 0.2
Set dash_gap, 0.1
# 6. Place the residue number on each cysteine for easy mapping
label (sulphurs), "%s%s" % (resn, resi)
Set label_color, Purple
Set label_size, 20
# 7. Set the transparent background for export and center the view
bg_color white
Set ray_opaque_background, 0
orient
```

Resources & Downloads

- SWISS-MODEL: https://swissmodel.expasy.org/
- PyMOL: https://pymol.org/

Troubleshooting

- If the chosen terminus turns out to be buried, the tag will block purification or the active site: use the opposite terminus and the matching vector.
- If a disulphide-dependent enzyme is expressed in a standard reducing strain it will misfold: move it to an oxidising-cytoplasm strain before repeating the expression.